# Understanding Student Performance in Secondary-School Mathematics

Final Project, Scenario 2 | Data Analytics and Statistics with Applications (LEAP)
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**Research question.** Which student background and behavioural characteristics are associated with final Mathematics performance, which factors are linked to the risk of failing, and what are the realistic limits of these observational data?

**Summary.** Across both linear and logistic specifications, past academic failure is the single strongest predictor of poor final performance and course failure. Higher maternal education correlates with better outcomes, while frequent socialising is linked to lower marks. Still, the models capture under a fifth of the total variance in grades. Importantly, the 38 students recorded with a grade of zero appear to be non-attendees or dropouts rather than test-takers who earned zero; handling them separately substantially shifts several estimates, particularly the effect of absences. Given the cross-sectional, observational structure, these results indicate associations rather than causal effects.

## 1. Data preparation and screening

The analysis uses the Mathematics subset of the UCI Student Performance dataset (Cortez, 2008), comprising 395 students across two Portuguese secondary schools: Gabriel Pereira (349) and Mousinho da Silveira (46). The target outcome is the final grade, G3, evaluated on a 0–20 scale, alongside a binary pass indicator (G3 ≥ 10). Following the project brief, earlier term marks (G1 and G2) were excluded from the predictor pool to focus on broader student background and behaviour.

Categorical survey items with natural ordering (such as parental education levels from 0 to 4, study time, and self-reported health or social habits on 1–5 scales) were retained as numeric integers for baseline modelling, while purely nominal fields were set as factors in R.

The dataset contains no missing cells, but inspection revealed an anomaly: 38 students (9.6%) have a final grade of exactly 0, with a complete void between 1 and 3. All 38 have zero recorded absences (compared to 22% zero-absence rates across the rest of the cohort), and their mean first-period grade was 7.5. This pattern suggests administrative dropouts or exam absentees whose attendance tracking simply ceased, rather than students who sat the exam and scored zero. Because abandoning the course still constitutes failing in practice, I retain them in the main models but evaluate the stability of the conclusions via a sensitivity check without them.

From an ethics standpoint, while individual names are omitted, the survey touches on sensitive areas including home stability, substance use, and personal relationships. Because the smaller school has only 46 records, cross-referencing demographic variables could lead to re-identification. The findings are intended to highlight broad patterns where academic assistance might help, not to profile or categorise individuals.

## 2. Exploratory data analysis

The average final grade is 10.4 with a standard deviation of 4.6 (median 11, IQR 8–14). Excluding the cluster at zero, the distribution is roughly bell-shaped around 11 (Figure 1). Overall, 67.1% of the cohort passed (265 students) and 32.9% failed (130 students), with the 38 zero-grade cases accounting for nearly thirty percent of all failures. Absences show marked right-skew (mean 5.7, median 4, reaching up to 75). Most participants had zero previous course failures (312 students), whereas 50 had one, 17 had two, and 16 had three or more. The gender split is fairly even (208 female, 187 male), whereas ambition for higher education is heavily skewed towards 'yes' (375 vs 20).

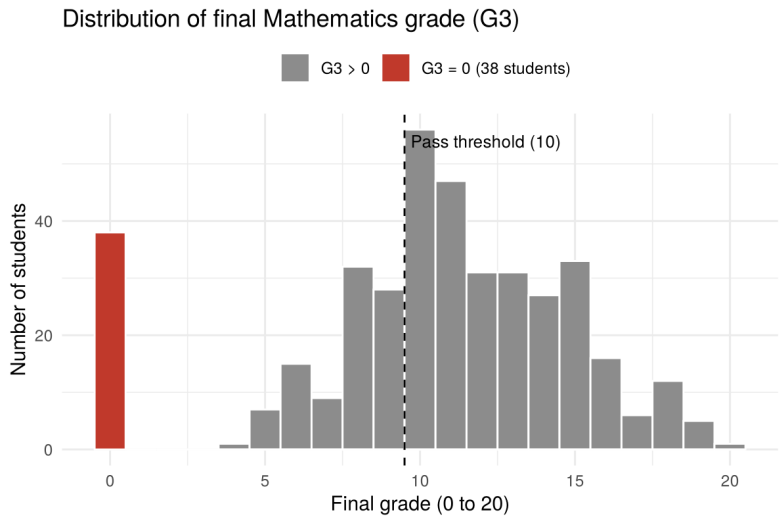
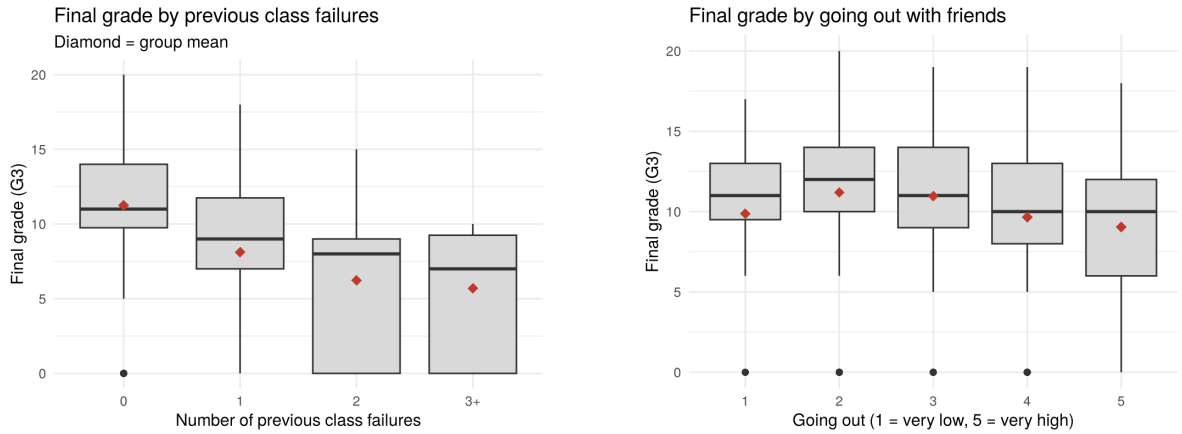



**Figure 1:** Distribution of final mathematics grades, highlighting the cluster of 38 non-attendees at zero.

Performance differences across groups are substantial. Average grades drop progressively from 11.3 among students with clean academic records down to 8.1, 6.2, and 5.7 for those with one, two, or three-plus past failures (Figure 2a). Socialising displays a slight non-linear trend: students who report going out very rarely score somewhat below average, those reporting low social outings peak at an average of 11.2, and those at the top end of the scale drop to 9.0 (Figure 2b). Male students averaged slightly higher than female students (10.9 vs 10.0), with an associated pass rate of 70.6% compared to 63.9%.




(a) By previous failures

(b) By going out frequency

**Figure 2:** Grade distributions across key student characteristics (box: IQR, diamond: mean).

## 3. Bivariate associations and variable choice

Rank-order (Spearman) correlations largely mirrored Pearson coefficients across the ordinal indicators (full matrix in Appendix A). Past failure count shows the clearest negative relationship with final score (r = -0.36), followed by maternal education (r = 0.22), age (-0.16), paternal education (0.15), and going out frequency (-0.13). Weekly study time correlates modestly (r = 0.10), while raw absences appear almost completely detached from grades overall (r = 0.03). Parental education levels are strongly collinear (r = 0.62), as are weekday and weekend drinking (r = 0.65); consequently, only one variable from each pair was considered for modelling to avoid redundancy.

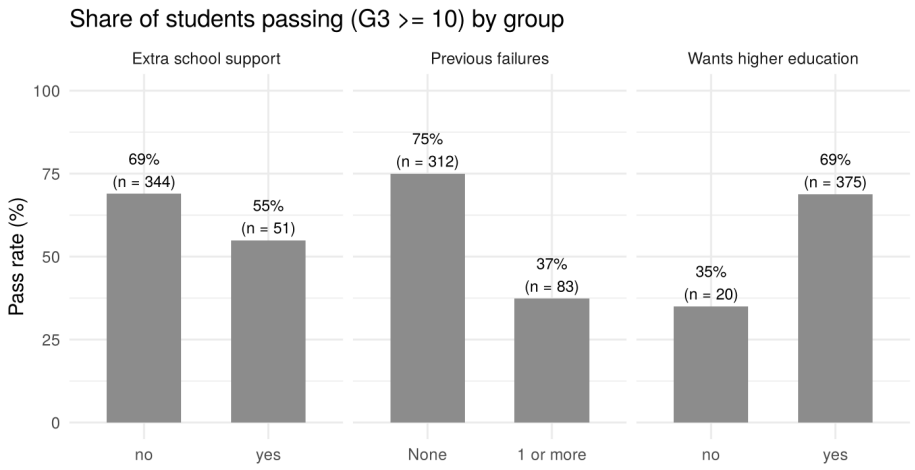



**Figure 3:** Pass rates (G3 ≥ 10) across selected binary indicators.

Group comparisons highlight similar divides. Students with prior failures pass at roughly half the rate of those without (37.3% vs 75.0%; $\chi^2 = 40$, $p < 0.001$). The pass rate among students not aiming for higher education is 35% compared to 69% for those who do ($p = 0.003$), though this group comprises only 20 individuals. Counterintuitively, students enrolled in remedial school support pass less often than those who are not (54.9% vs 68.9%, $p = 0.07$). This most likely reflects selection rather than an adverse effect of tutoring; support is allocated to students who are already struggling. For this reason, schoolsup was left out of the predictive regressions.

The apparent lack of relationship between absences and performance is driven by the 38 zero-grade students, all recorded with zero absences at the bottom-left of the scatter (Figure 4b). Once these 38 cases are removed, the trend between absences and grades turns negative, suggesting that the overall flat correlation is an artefact of data logging. For the final models, I retained past failures, maternal education, going out, and sex as core background factors, and added study time, absences, and higher-education aspiration.

## 4. Regression analysis

The linear specification models G3 against the seven selected variables, using female and no aspiration for higher education as reference levels (Table 1).

**Table 1:** Linear regression model for final Mathematics grade (n = 395).

| Predictor | Estimate | 95% CI | t | p |
|---|---|---|---|---|
| Previous failures | -1.80 | [-2.41, -1.20] | -5.85 | <0.001 |
| Mother's education | 0.49 | [0.09, 0.89] | 2.43 | 0.016 |
| Going out | -0.46 | [-0.83, -0.08] | -2.38 | 0.018 |
| Study time | 0.37 | [-0.15, 0.90] | 1.39 | 0.164 |
| Absences | 0.04 | [-0.02, 0.09] | 1.40 | 0.162 |
| Male | 1.41 | [0.52, 2.30] | 3.13 | 0.002 |
| Wants higher education | 1.78 | [-0.24, 3.80] | 1.73 | 0.084 |

*$R^2 = 0.186$, Adjusted $R^2 = 0.171$, Residual SD = 4.17 points, n = 395.*

Holding the other predictors constant, each prior failure is associated with a final grade about 1.8 points lower on the 20-point scale. A student with average profile values (secondary-educated mother, moderate social life, 2–5 hours study, 4 absences, female, aiming for university) has an expected score of 10.5 with no failures, dropping to 6.9 with two previous class failures. Each level of maternal education is associated with roughly half a point more, while each step up the social outing scale is associated with about half a point less. Male students average 1.4 points higher when other predictors

are held equal. Overall, the fit remains modest ($R^2 = 0.186$); more than four-fifths of grade variance remains uncaptured, and predictions carry an average uncertainty of over 4 marks.

**Table 2:** Logistic regression model for passing final Mathematics (n = 395).

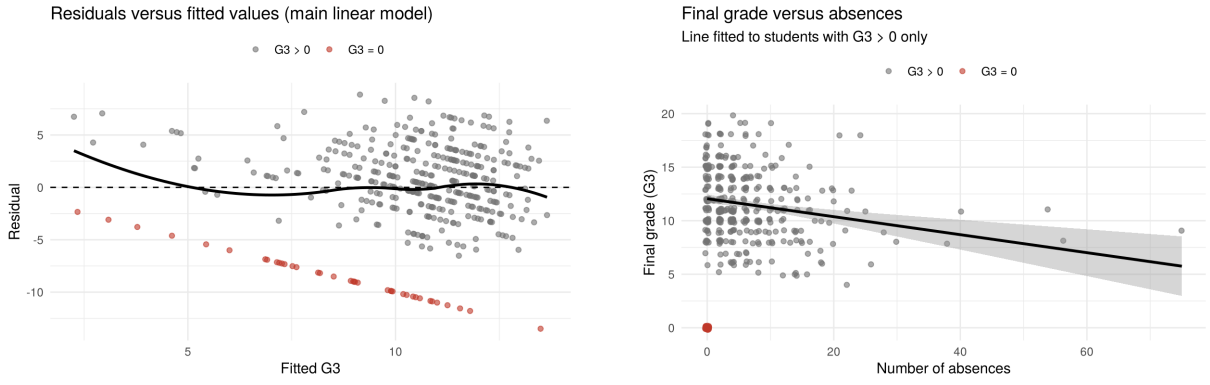
| Predictor | Odds ratio | 95% CI | p |
|---|---|---|---|
| Previous failures | 0.42 | [0.29, 0.59] | <0.001 |
| Mother's education | 1.11 | [0.89, 1.38] | 0.368 |
| Going out | 0.71 | [0.57, 0.87] | 0.001 |
| Study time | 1.09 | [0.81, 1.48] | 0.559 |
| Absences | 0.98 | [0.96, 1.01] | 0.244 |
| Male | 1.74 | [1.06, 2.89] | 0.030 |
| Wants higher education | 2.13 | [0.71, 6.62] | 0.177 |

The logistic model mirrors these patterns on the pass/fail margin (Table 2). Holding the other predictors constant, each past failure is associated with roughly 58% lower odds of passing (OR = 0.42), and each unit increase in going out with 29% lower odds (OR = 0.71). Converted to probabilities for the baseline profile, a student with zero failures has a 72% pass chance, which falls to 31% with two past failures. Interestingly, maternal education shows no clear link to the passing threshold itself (p = 0.368), despite its positive slope in the linear model. As a standalone classifier with a default 0.5 cut-off, the logistic model achieves 72.9% overall accuracy, which is only slightly above the naive baseline of predicting a pass for everyone (67.1%), successfully flagging only 35% of actual course failures.

The two models answer different questions. The linear model describes how far grades differ across the whole 0 to 20 range, while the logistic model addresses the outcome that matters most for a school, the risk of failing, and allows results to be expressed as probabilities. The contrast for maternal education shows why both are useful: it is associated with grade level, but not clearly with the chance of crossing the pass mark.

## 5. Diagnostics and sensitivity analysis

Residual plots from the linear fit reveal a distinct structural problem: the 38 zero-grade records line up in a clean diagonal trace well below the main scatter (Figure 4a). Beyond this subgroup, residuals remain reasonably balanced around zero, aside from some boundary compression at the low end. The Q-Q plot (Appendix A) confirms this substantial left-tail distortion. Multicollinearity is negligible, with variance inflation factors between 1.03 and 1.19.




(a) Residuals vs fitted values

(b) Absences vs final grade

**Figure 4:** Residual behaviour and the impact of the non-attending cohort (marked red).

Statistical uncertainty differs considerably across predictors. In the linear model, past failures is the most convincing result: its 95% confidence interval (-2.41 to -1.20) lies far from zero (t = -5.85). Maternal education and going out are significant at the 5% level, but their intervals come close to zero (0.09 to 0.89 and -0.83 to -0.08), so the true associations could be very small. The higher-education

coefficient is large (1.78) but imprecise (-0.24 to 3.80, p = 0.084), because only 20 students do not aspire to university. In the logistic model, the odds ratios for past failures (0.29 to 0.59) and going out (0.57 to 0.87) both clearly exclude 1, the interval for sex (1.06 to 2.89) only just does, and the interval for higher education (0.71 to 6.62) is too wide to support any conclusion.

Statistical significance and practical importance do not always coincide. A difference of 1.8 points per failure is large in practice, since it can move a student from a pass to a fail. The half-point differences linked to going out and maternal education are statistically detectable but small relative to a typical prediction error of about 4 points. Conversely, the higher-education gap may matter in practice even though these data cannot estimate it precisely.

Excluding these 38 cases in a sensitivity run substantively alters several parameters (Appendix C). Most strikingly, the coefficient for absences flips from slightly positive to clearly negative (-0.07 points per day missed, 95% CI [-0.10, -0.03], p < 0.001). The penalty associated with past failures is cut roughly in half (from -1.80 to -0.98), reflecting that non-completers carried a disproportionate record of prior failures (0.92 vs 0.27 on average). Study time turns marginally significant (0.42, p = 0.04), while the higher-education coefficient collapses and the male advantage weakens. Meanwhile, the estimated effects for maternal education and going out remain virtually identical, suggesting that their associations do not depend on how dropouts are handled.

Two further cautions apply. Ordered survey items enter the models as linear scores, although the going-out pattern is visibly non-linear, so a single slope is a simplification. The predictors were also chosen after exploring the same data, which makes the p-values somewhat optimistic, and no out-of-sample validation was carried out.

## 6. Discussion and limitations

Because these data reflect an observational snapshot of two public secondary schools in 2005–2006, the associations cannot be interpreted causally. One cannot conclude that simply limiting a student's social life would directly produce higher test scores.

Unobserved factors are likely doing much of the explanatory work. Prior foundational math competence, individual drive, household income, teacher effectiveness, and peer dynamics were not tracked. Past failures are most likely an indicator of cumulative academic struggle rather than an independent causal mechanism. Similarly, self-reported measures on brief integer scales carry unavoidable noise and reporting bias.

The findings may also not generalise. The sample comes from two public schools in one region of Portugal, with 88% of students from a single school, in one school year and one subject. Grading scales, repetition policies and support systems differ across schools, countries and decades, so the size, and even the direction, of some associations could differ elsewhere.

In summary, past academic difficulty remains the most reliable warning sign for poor final performance in this dataset, with family background and social balance playing secondary, consistent roles. However, the data leave over 80% of student achievement unexplained, and several baseline conclusions, notably the role of absences, are heavily skewed by how non-attending students are handled. The model provides a broad overview of risk factors across groups, but it lacks the precision needed to guide individual interventions.

Several claims would go beyond this evidence: that failing a class, going out or a mother's education causes lower grades; that boys are inherently better at mathematics; that extra school support is ineffective; that absences do not matter; or that the model can reliably identify which individual students will fail.

### References

Cortez, P. (2008). Student Performance [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5TG7T

Cortez, P., & Silva, A. (2008). Using data mining to predict secondary school student performance. In Proceedings of the 5th Future Business Technology Conference (FUBUTEC 2008), 5–12.

R Core Team (2026). R: A language and environment for statistical computing (version 4.6.1). Packages: ggplot2, dplyr, broom.

# Appendix A. Correlation heatmap and Q-Q plot

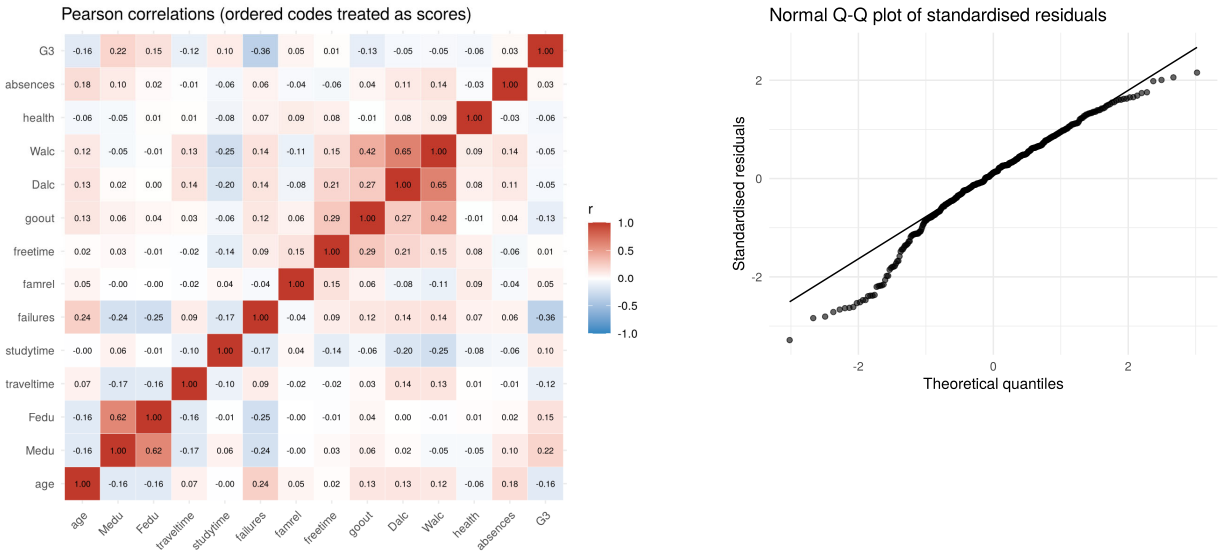



**Figure 5:** Left: Correlation matrix. Right: Normal Q-Q plot of standardised residuals from the full linear fit.

# Appendix B. Logistic model predictions

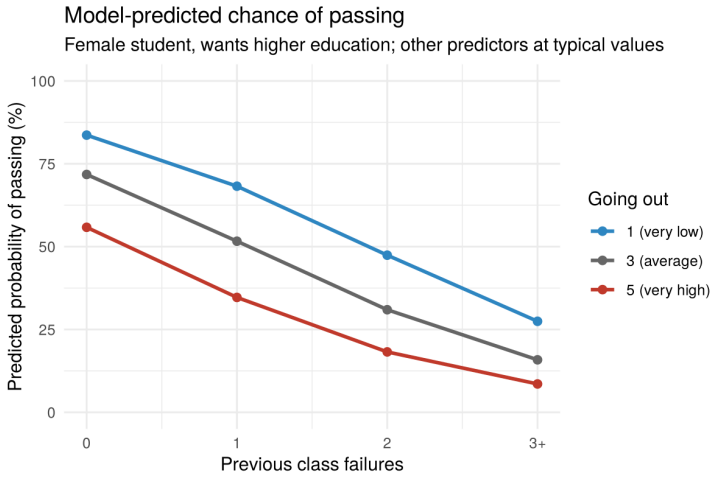



**Figure 6:** Predicted pass probabilities across past failures and social outings for a female student aspiring to higher education.

# Appendix C. Sensitivity analysis: all students vs excluding G3 = 0

| Predictor | Linear: All (n=395) | Linear: G3 > 0 (n=357) | Logistic: All (n=395) | Logistic: G3 > 0 (n=357) |
|---|---|---|---|---|
| Failures | -1.80 [-2.41, -1.20] | -0.98 [-1.47, -0.48] | 0.42 [0.29, 0.59] | 0.51 [0.35, 0.74] |
| Mother's educ. | 0.49 [0.09, 0.89] | 0.46 [0.17, 0.76] | 1.11 [0.89, 1.38] | 1.10 [0.85, 1.42] |
| Going out | -0.46 [-0.83, -0.08] | -0.46 [-0.75, -0.18] | 0.71 [0.57, 0.87] | 0.65 [0.51, 0.83] |
| Study time | 0.37 [-0.15, 0.90] | 0.42 [0.02, 0.81] | 1.09 [0.81, 1.48] | 1.13 [0.80, 1.60] |
| Absences | 0.04 [-0.02, 0.09] | -0.07 [-0.10, -0.03] | 0.98 [0.96, 1.01] | 0.95 [0.92, 0.98] |
| Male | 1.41 [0.52, 2.30] | 0.81 [0.16, 1.47] | 1.74 [1.06, 2.89] | 1.49 [0.86, 2.63] |
| Wants higher ed. | 1.78 [-0.24, 3.80] | 0.09 [-1.60, 1.79] | 2.13 [0.71, 6.62] | 1.40 [0.38, 5.02] |
| $R^2$ (linear) | 0.186 | 0.180 | | |